Chapter revision — angles in polygons

The whole of Chapter I pulled together, in the shape of Cambridge Project 2, "Angle tangle".

LESSON 17 | 1 × 40 MIN | GEOMETRY 8, ТЕСТ 1 | STAGE 9 · PROJECT 2 [CAMBRIDGE INSERT]

LESSON CLOCK

4' | 10' | 8' | 16' | 2'

Recall round	4 min
Project 2	10 min
Interactive	8 min
Mixed practice	16 min
Homework	2 min

LEARNING OBJECTIVES

- Recall the definitions, properties and tests of Chapter I without notes.
- Choose the right property for a given problem.
- Work the Cambridge "Angle tangle" investigation.
- Identify personal gaps before the control work.

TEXTBOOK

National	Тест 1, p. 33
Cambridge	Learner's Book p. 127
Workbook	Workbook 5.2–5.3

01 Explanation

The recall round — four minutes, no notes

Ask these in order, round the class. Anything that stalls is what tomorrow's control work will catch.

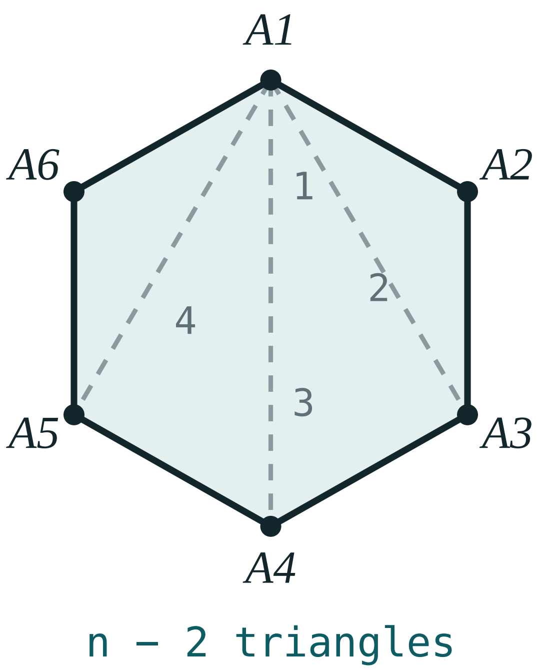
1. Sum of the interior angles of an n-gon? $180°(n-2)$
2. Sum of the exterior angles? $360°$, always.
3. Three properties of a parallelogram? Opposite sides equal, opposite angles equal, diagonals bisect each other.
4. Three tests for a parallelogram? One pair equal and parallel; both pairs of sides equal; diagonals bisect each other.
5. What is special about the diagonals of a rectangle? They are equal.
6. Of a rhombus? Perpendicular, and they bisect the angles.
7. Of a square? All of the above.
8. Midline of a triangle? $MN \parallel BC$, $MN = \dfrac{1}{2}BC$.
9. Midline of a trapezium? $\dfrac{a+b}{2}$.
10. Thales' theorem? Parallel lines cut proportional segments on any transversals.

Project 2 · Angle tangle

The Cambridge investigation, adapted to the national chapter. Work in pairs.

THE TASK

1. Draw any convex pentagon and measure its five interior angles. Add them. Compare with $180° \cdot 3 = 540°$.
2. Now draw a **non-convex** pentagon and repeat. Does the formula still hold?
3. Extend each side in turn and measure the exterior angles. What do they add to?
4. Predict the answers for a 9-gon *before* drawing it. Then check.
5. Write one sentence explaining why the exterior sum does not depend on n.



The diagonals from one vertex are what turn the polygon into triangles — the heart of the whole investigation.

Expect the surprise at step 2: the interior formula survives for non-convex polygons (with reflex angles counted properly), but the exterior-angle argument does not, because “turning through the exterior angle” goes backwards at a reflex vertex.

The three traps to name out loud

BEFORE THE CONTROL WORK

1. **“The diagonals of a parallelogram are equal.”** They bisect each other. Equal needs a rectangle.

2. **“Equal diagonals mean a rectangle.”** Only if the shape is already a parallelogram — an isosceles trapezium has equal diagonals too.
3. **Mixing part-to-part with part-to-whole** in a Thales proportion. Decide which one and stay with it on both sides.

02 Worked examples

EXAMPLE 1 **A regular polygon has interior angle 150°. Find n, and the sum of its interior angles.**

1. $exterior = 180° - 150° = 30°$

   Work with the exterior angle.

2. $n = \dfrac{360°}{30°} = 12$

3. $sum = 180°(12 - 2) = 1800°$

   Check with the interior formula.

ANSWER

$n = 12$, sum $1800°$

EXAMPLE 2 ***ABCD is a parallelogram with AC = BD. What kind of quadrilateral is it, and why?***

1. It is already a parallelogram, so the diagonals bisect each other.
2. Equal diagonals plus bisecting gives four equal half-diagonals.
3. $\triangle ABC \cong \triangle BAD$ by SSS, so $\angle A = \angle B$.
4. They are supplementary, so each is $90°$ — a rectangle.

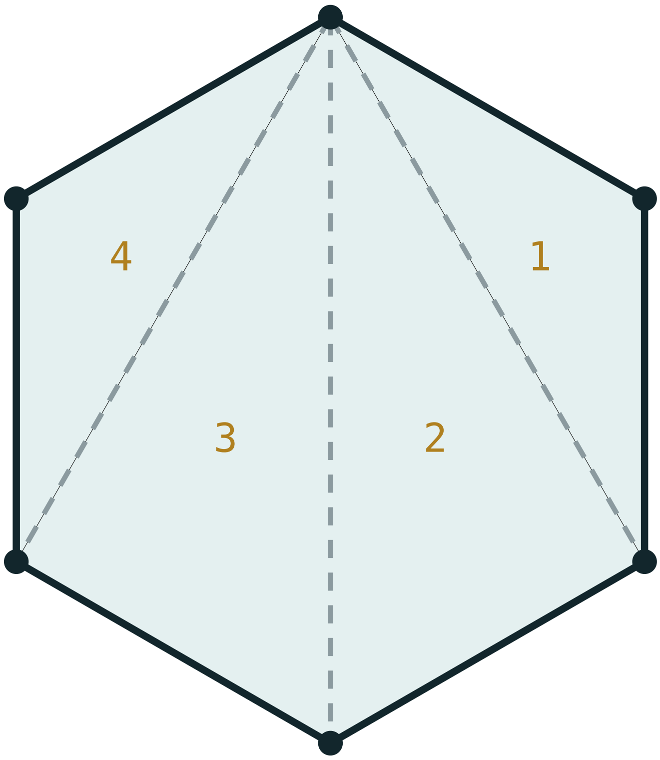
ANSWER

A rectangle.

03 Interactive model

Run the polygon model up to 12 sides while the class checks their Project 2 predictions.

Angle sum of a polygon



$$180° \times (6 - 2) = 720°$$

sides n **6** | triangles **4** | interior sum **720°** | each angle **120°** | exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Investigation	Tadqiqot	Исследование
Predict	Bashorat qilish	Предсказать
Reflex angle	Yoyiq burchakdan katta burchak	Рефлексный угол
Justify	Asoslash	Обосновать
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Family tree	Tasnif sxemasi	Схема классификации

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A regular polygon has interior angle 150°. It has:

- 10 sides
- 12 sides
- 15 sides
- 20 sides

A parallelogram with equal diagonals is:

- a rhombus
- a rectangle
- a trapezium
- a kite

A parallelogram with perpendicular diagonals is:

- a rhombus
- a rectangle
- a trapezium
- not determined

The midline of a trapezium with bases 9 and 15 is:

- *6*
- *12*
- *24*
- *3*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Sum of the interior angles of a nonagon (9 sides).*

   ANSWER *$1260°$*

2. *Each exterior angle of a regular octagon.*

   ANSWER *$45°$*

3. *In parallelogram ABCD, $\angle A = 55°$. Find $\angle B$.*

   ANSWER *$125°$*

4. *The diagonals of a rhombus are 8 and 6. Find the area.*

   ANSWER *24*

5. *A rectangle has sides 5 and 12. Find the diagonal.*

   ANSWER *13*

6. *A trapezium has bases 9 and 15. Find the midline.*

   ANSWER *12*

7. *In $\triangle ABC$, $BC = 16$. Find the midline parallel to BC.*

   ANSWER *8*

Medium · the standard question

7 PROBLEMS

1. *A regular polygon has interior angle 150°. Find n and the interior sum.*

   ANSWER $n = 12$, $1800°$

2. *ABCD is a parallelogram with AC = BD. Name the shape and justify.*

   ANSWER A rectangle — equal and bisecting diagonals force right angles.

3. *A rhombus has side 10 and one diagonal 12. Find the other diagonal and the area.*

   ANSWER 16, area 96

4. *An isosceles trapezium has bases 6 and 16 and legs 13. Find the height and area.*

   ANSWER $h = 12$, $S = 132$

5. *$MN \parallel BC$ with $AM = 4$, $MB = 8$ and $AN = 5$. Find NC.*

   ANSWER 10

6. *The angles of a quadrilateral are in the ratio 3 : 4 : 5 : 6. Find them.*

   ANSWER $60°$, $80°$, $100°$, $120°$

7. *A square has diagonal 12. Find its area.*

   ANSWER 72

Hard · stretch and reason

7 PROBLEMS

1. *Project 2, step 4: predict the interior and exterior sums for a 9-gon, then check.*

   ANSWER $1260°$ interior, $360°$ exterior.

2. *Project 2, step 5: explain in one sentence why the exterior sum does not depend on n.*

   ANSWER Walking once round the polygon turns you through exactly one full revolution, whatever the number of corners.

3. *Prove that the midpoints of the sides of a rhombus form a rectangle.*

   ANSWER Each side of the new quadrilateral is parallel to a diagonal, and the diagonals of a rhombus are perpendicular — so the new angles are right angles.

4. *A parallelogram has perimeter 40 and one diagonal 16, with the diagonals perpendicular. Find the sides and the other diagonal.*

   ANSWER Perpendicular diagonals make it a rhombus: side 10, half-diagonals 8 and 6, so the other diagonal is 12.

5. *In trapezium ABCD (AD ∥ BC), the midline is 10 and the height is 7. A diagonal splits it into two triangles — find their total area.*

   ANSWER $S = 10 \cdot 7 = 70$ — the two triangles together are the whole trapezium.

6. *A convex polygon has exactly three right angles and the rest equal. If n = 6, find the equal angles.*

   ANSWER $720° - 270° = 450°$ shared by 3 angles: $150°$ each.

7. *Explain why a non-convex polygon can break the exterior-angle argument.*

   ANSWER At a reflex vertex the turn is in the opposite direction, so the turns no longer add to a single full revolution.

07 Homework

Homework before the control work

Geometry 8, Тест 1, p. 33. Then revise every definition and property from Темы 1–11.

1. *Work Тест 1 on p. 33 in full.*

2. *Write out the three tests for a parallelogram, with a sketch each.*
3. *Write out the diagonal properties of the parallelogram, rectangle, rhombus and square in one table.*
4. *A regular polygon has interior angle 162°. Find n.*
5. Finish Project 2 and write your one-sentence explanation for step 5.